

FINAL LAB PROJECT

Master in Data Science — Machine Learning (UPC - FIB)

1. Objective

The goal of this project is to design, implement, and evaluate a complete Machine Learning pipeline on a real-world dataset.

The project must integrate multiple techniques covered in the course, including:

- Unsupervised Learning
- Linear and Regularized Models
- Kernel Methods
- Ensemble Methods

This project is designed to help you connect the different techniques seen in class into a single coherent pipeline.

2. General Information

- The project must be completed in pairs (2 students)
- Implementation must be done in Python (e.g., *scikit-learn*, *numpy*, *pandas*)
- All results must be reproducible (use fixed random seeds where needed)
- Any external resources (code, ideas, libraries) must be properly cited

3. Dataset Selection

You must choose one of the following datasets, or propose a similar one (subject to instructor approval):

Available options

Option A — Online News Popularity (Regression / Classification)

Predict article popularity (number of shares)

Option B — Communities and Crime (Regression)

Predict violent crime rates (high-dimensional dataset with missing values)

Option C — Bank Marketing (Classification)

Predict whether a client subscribes to a term deposit (categorical features and imbalance)

Option D — Credit Card Default (Classification)

Predict default payments (financial dataset with strong predictive signal)

Dataset Requirements

Your chosen dataset must:

- Contain numerical and/or categorical variables
- Be a real dataset (not synthetically generated)
- Contain more than 10 features
- Contain more than 200 samples
- Require non-trivial preprocessing
- Be sufficiently documented to interpret results
- Not be a trivial dataset (e.g., Iris, MNIST)

4. Project Philosophy

Your goal is not to obtain the highest possible performance.

Instead, you will be evaluated on:

- Your understanding of the data
- The methodology applied
- The quality of your analysis and reasoning

Different datasets have different levels of difficulty. Evaluation will focus on process, not raw performance.

5. Required Project Phases

Your project must include the following phases:

Phase I — Unsupervised Structural Analysis

- Apply PCA and analyze explained variance
- Perform clustering (K-Means and GMM)
- Justify the number of clusters (Elbow Method / Silhouette Score)
- Discuss whether clusters align with the target variable

Phase II — Linear and Regularized Models

- Train:
 - Logistic Regression (classification)
 - Ridge / Lasso (regression)
- Study the effect of regularization (λ)
- Analyze how model coefficients change

Phase III — Kernel Methods (SVM)

- Train SVM models using:
 - Linear kernel
 - Polynomial kernel
 - RBF kernel
- Perform hyperparameter tuning (Grid Search)

Phase IV — Ensemble Methods

- Train:
 - Random Forest (Bagging)
 - Boosting models (AdaBoost or Gradient Boosting)
- Compare:
 - Performance
 - Training time
 - Feature importance

Phase V — Model Selection and Validation

- Use cross-validation (preferably nested CV)
- Evaluate using appropriate metrics:

For classification:

- Accuracy
- F1-score
- ROC-AUC
- AUPR

For regression:

- R^2
- MAE
- RMSE
- Compare models and select the best one
- *(Optional)* Perform statistical tests (e.g., paired t-test)

Baseline Models (Mandatory)

You must include at least one simple baseline model to contextualize performance.

6. Required Analysis Components

All projects must include:

- Exploratory Data Analysis (EDA)
- Data preprocessing (missing values, encoding, scaling)
- Feature selection or extraction
- Model comparison
- Interpretation of results
- Discussion of dataset challenges

7. Deliverables

Project Proposal (1 page)

To be submitted at the beginning of the project:

- Team members
- Dataset chosen
- Problem definition
- Motivation
- Dataset description (samples, features, variable types)
- *(Optional)* related work

Final Submission

1. Technical Report (PDF)

- Maximum 12 pages
- Written in a scientific style (similar to a research paper)
- Must include:
 - Problem description
 - Methodology
 - Experiments
 - Results
 - Conclusions

2. Code

- Python notebooks or scripts
- Clean, modular, and reproducible

3. Reproducibility File

- Clear instructions on how to run the code

Code and report must be submitted separately

8. Evaluation Criteria

Projects will be graded based on:

- Appropriate use of ML techniques
- Rigor in evaluation (cross-validation, metrics, methodology)
- Quality of results (generalization, interpretability)
- Quality of the report (clarity, structure, explanations)

Special emphasis on:

- Insight into results
- Understanding of the data
- Justification of decisions

9. Key Dates

- **Project proposal:** *21st April*
- **Final submission:** *15th June*

Final Note

This project simulates a real-world machine learning workflow.

Simply applying models is not enough. You must analyze, justify, and interpret every step.